END YEAR NETWORKING PROJECT

Vic Modern Hotel – Network Design and Implementation
Problem Statement

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1. Project Overview

As part of the end year networking project, students are required to design and implement the network infrastructure for Vic Modern Hotel. The hotel consists of three floors, each hosting a distinct set of departments. The first floor accommodates the Reception, Store, and Logistics departments. The second floor hosts the Finance, Human Resources (HR), and Sales/Marketing departments. The third floor is designated for the IT and Administration (Admin) departments.

2. Network Requirements

The following requirements must be satisfied during the design and implementation of the network:

2.1 Router Configuration

Three routers shall be deployed to interconnect each floor. All routers are to be physically placed in the server room located in the IT department on the third floor.

2.2 Router Interconnection

All routers shall be connected to one another using Serial DCE cables to form a full-mesh topology between the three routers.

2.3 Inter-Router Network Addressing

The point-to-point networks between the routers shall use the following address ranges: 10.10.10.0/30, 10.10.10.4/30, and 10.10.10.8/30.

2.4 Switch Placement

Each floor shall be equipped with one managed switch, physically placed on its respective floor.

2.5 Wireless Connectivity

Each floor shall have a wireless access point providing Wi-Fi connectivity for laptops, smartphones, and other wireless-capable devices.

2.6 Printers

Each department shall be equipped with at least one printer connected to its local network.

3. VLAN and Network Addressing Plan

Each department shall be isolated in its own Virtual Local Area Network (VLAN). The VLAN identifiers and associated network addresses are defined as follows:

Floor	Department	VLAN ID	Network Address
1st Floor	Reception	VLAN 80	192.168.8.0/24
1st Floor	Store	VLAN 70	192.168.7.0/24
1st Floor	Logistics	VLAN 60	192.168.6.0/24

Floor	Department	VLAN ID	Network Address
2nd Floor	Finance	VLAN 50	192.168.5.0/24
2nd Floor	Human Resources (HR)	VLAN 40	192.168.4.0/24
2nd Floor	Sales / Marketing	VLAN 30	192.168.3.0/24
3rd Floor	Administration (Admin)	VLAN 20	192.168.2.0/24
3rd Floor	Information Technology (IT)	VLAN 10	192.168.1.0/24

4. Routing Protocol

Open Shortest Path First (OSPF) shall be used as the dynamic routing protocol across all three routers. OSPF must be configured to advertise all directly connected networks and ensure full reachability between all departments and floors.

5. DHCP Configuration

All end devices within the network shall obtain their IP addresses dynamically via DHCP. Each router shall act as the DHCP server for the VLANs hosted on its respective floor, providing automatic address assignment to all connected hosts.

6. Network Communication

All devices across all floors and departments must be able to communicate with one another. Full inter-VLAN and inter-floor connectivity must be verified as part of the implementation.

7. Remote Access – SSH Configuration

Secure Shell (SSH) shall be configured on all three routers to enable encrypted remote management access. SSH must be functional and tested from an authorized host within the network.

8. Test PC – IT Department

A dedicated PC named Test-PC shall be added to the IT department and connected to port fa0/1 of the IT department switch. This device shall be used to verify and test SSH remote login functionality to the routers.

9. Port Security

Port security shall be configured on the IT department switch for port fa0/1 to restrict access exclusively to the Test-PC. The sticky MAC address learning method shall be used to dynamically record the authorized MAC address. The violation mode shall be set to shutdown, meaning the port will be automatically disabled upon detection of an unauthorized device.

This document constitutes the official problem statement for the end year networking project. All requirements listed herein are mandatory and must be fully implemented and demonstrated.